

Hamza N. Hassan

Software Engineer · Computer Engineering

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SUMMARY

Software engineer building multichannel messaging and marketing-automation features for a WhatsApp Business and e-commerce SaaS platform, with prior experience in banking and payment systems (SWIFT MT and ISO 20022). Works across the stack with Kotlin, Java, Spring Boot, PostgreSQL, and React, with a focus on clean, well-tested code.

WORK EXPERIENCE

Software Engineer · **BusinessChat**

Jun 2026 – Present

FULL-TIME

Multichannel Inbox

- Integrated **Instagram, Facebook Messenger, and Telegram** into the shared inbox end to end: account connection and provisioning, webhook verification, and inbound and outbound messaging.
- Delivered rich messaging features across channels: media and albums, reactions, message edits, delivery and read statuses, typing indicators, ice breakers, and persistent menus.
- Refactored inbound message handling into a **single pipeline with one adapter per channel** (WhatsApp, Instagram, Messenger, TikTok, Telegram, LiveChat), unifying deduplication, contact resolution, and latency telemetry.

Marketing Automation & WhatsApp

- Built **webhook-triggered automations** with filtering rules, per-filter variable mappings, and configurable request security.
- Shipped automations for OTP verification with WhatsApp authentication templates, CSAT feedback, purchase reviews, and e-invoices.
- Switched outbound WhatsApp audio to native **voice messages** with an OGG/Opus transcoding path, and fixed template-send failures.
- Extended billing with prepaid plans and per-channel service-message pricing.

Platform & Engineering

- Built **event-driven services on Google Cloud**: publishing and consuming Protobuf events over **Pub/Sub** with backward-compatible schema changes, deployed on **Cloud Run**.
- Used **Redis** for caching, deduplication, rate limiting, and feature flags in a multi-tenant SaaS platform.
- Modelled data in **PostgreSQL** with Flyway migrations and type-safe **jOOQ** queries, and fed analytics and reporting views in **BigQuery**.
- Wrote non-blocking Kotlin services with **coroutines** on reactive Spring, and made webhook and message processing reliable with idempotency, retries, rate limits, circuit breakers, and HMAC signature checks.
- Added tracing and latency telemetry, backed changes with integration tests (WireMock, real databases), and kept OpenAPI-generated clients in sync for web and iOS.
- Worked across an ongoing **Scala-to-Kotlin migration**, and used an AI-assisted workflow (Claude Code) for planning, implementation, and code review.

Tech: Kotlin (coroutines), Spring Boot, Scala, React, TypeScript, PostgreSQL, jOOQ, Flyway, Redis, Google Cloud (Pub/Sub, Cloud Run, BigQuery), Protobuf, WireMock, Meta Graph API, Telegram Bot API

Software Engineer · ProgressSoft

Sep 2024 – May 2026

FULL-TIME

PayHub

- Core team member on **PayHub**, a platform for banking and payment integrations.
- Implemented **SWIFT MT message structural validation** and financial message parsing according to SWIFT standards.
- Developed backend services and **RESTful APIs** with Java and Spring Boot for financial transaction processing, integrated with Domino-based frontends.
- Implemented XML and message-format validations (reference matching, ISO 8601 dates, message status verification).
- Wrote unit and integration tests with JUnit, Mockito, WireMock, and RestAssured; worked with PostgreSQL and MySQL.
- Contributed to CI/CD pipelines using GitLab, Docker, and Kubernetes, following Clean Code and Agile Scrum practices.

Payment Messages Agentic Workflow

- Contributed to a proof of concept for an **AI-driven coding agent** to automate financial message processing and replace core Provide Core functionality.
- Designed an agentic workflow for **bidirectional MT ↔ MX conversion** between SWIFT MT and ISO 20022 MX messages, including mapping and generation logic.
- Developed AI-assisted validation pipelines for both MT and MX messages, implemented entirely in Kotlin.

Software Developer · Leading Point

Apr 2024 – Sep 2024

FULL-TIME

- Led development of an **automation API with Python and FastAPI**, significantly reducing manual effort for the process team.
- Streamlined data retrieval from Azure DevOps, SharePoint, and SonarQube.
- Developed **Q-POINT**, a Django-based process management system with an HTML, CSS, and JavaScript frontend, and managed its PostgreSQL databases.
- Fixed bugs and designed new pages in React for the Vision Point project.

Software Engineering Intern · Wiley / Atypon

May 2023 – Oct 2023

INTERNSHIP

- Implemented a **decentralized, cluster-based NoSQL database** using Java, Spring Boot, and Docker.
- Built a web app with Sockets, Servlets, and Spring Boot to compare them and show how Java web technology evolved.
- Created an extensible **Uno game engine** using OOP and design patterns, and a **multithreaded Old Maid card game** where each player runs as a thread.
- DevOps: Git and GitHub workflows, Docker-based microservices, and Linux shell scripts for file-analysis reports.

EDUCATION

Bachelor of Engineering, Computer Engineering · University of Jordan

Sep 2019 – Jan 2024

Amman, Jordan · High school average: 95.5

PROJECTS

Decentralized Cluster-Based NoSQL Database

github.com/hamzahassanocode/Decentralized-Cluster-Based-NoSQL-Database

- Decentralized NoSQL database in Java and Spring Boot with its own REST query API, focused on load balancing, data consistency, and efficient queries.
- Custom bootstrapping node for cluster initialization and user-node assignment; each node holds a replica in its own Docker container, with no central point of failure.
- School registration demo app; built with SOLID principles, Clean Code, and thorough testing.

Water Pump Automation App

github.com/hamzahassanocode/Flutter-App-Water-pump-Automation

- Flutter and Dart app integrated with Firebase and Arduino to monitor tank water levels and solar-cell performance.
- Remote pump and solar control with manual and automatic modes, plus requesting a refill from nearby water tankers.

Cardio Disease Prediction AI

github.com/hamzahassanocode/Cardio-Disease-Prediction-AI

- Analyzed a Kaggle dataset of 70,000 records and 13 features; trained RandomForest, SVC, and XGBoost models.
- Fine-tuned a Keras model reaching 72% accuracy and 88% recall at a 0.3 threshold.

Home Decor Shop

github.com/hamzahassanocode/Home-Decor-Shop-HTML-CSS-JS

- Responsive storefront built with HTML, CSS, and JavaScript, designed for seamless navigation across devices and screen sizes.

LICENSES & CERTIFICATIONS

- **ICPC Jordanian Collegiate Programming Contest (2021):** 11th place locally
- **Software Engineering Using Java and DevOps:** Wiley / Atypon training certificate
- **The Complete 2022 Flutter & Dart Development Course:** Udemy

TECHNICAL SKILLS

Languages	Kotlin, Java, Scala, Python, TypeScript, JavaScript, Dart, C++
Backend & APIs	Spring Boot, Kotlin Coroutines, RESTful APIs, Webhooks, Event-driven architecture, Protobuf, OpenAPI, FastAPI, Django, Servlets
Messaging & Payments	WhatsApp Business API, Instagram & Messenger APIs, Telegram Bot API, SWIFT MT, ISO 20022 MX
Frontend & Mobile	React, HTML, CSS, Flutter
Data	PostgreSQL, MySQL, Redis, BigQuery, NoSQL, H2, jOOQ, Flyway, Liquibase
Cloud & DevOps	Google Cloud (Cloud Run, Pub/Sub, BigQuery), Docker, Kubernetes (MicroK8s), CI/CD (GitLab), Observability, Git, Linux
Testing	JUnit, AssertJ, Mockito, WireMock, RestAssured, integration testing